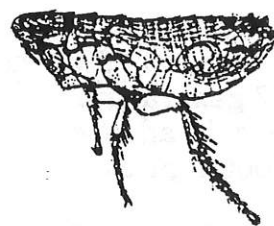


flea NEWS

38



Lie Chi-ying

Department of Entomology Iowa State University Ames, Iowa 50011-3222

The following was prepared by Professor Wang D. -q. and Doctor Wu H. -y. We are grateful for this contribution to our knowledge of this great scientist. In behalf of the readers of Flea News, we present our sincere condolences.

Liu Chi-ying

24 - III - 1905

14 - X - 1988

Professor Liu Chi-ying was born in Wuxian, Jianxi Province, China. He graduated from Nanking University in 1929 and became a member of Phi Tau Phi Honorary Society of the United States. In 1933 he entered the graduate program of the Department of Entomology and Economic Zoology at the University of Minnesota where he earned an MS degree under the tutorship of Dr. William A. Riley. He was elected a member of both the Sigma Xi and Gamma Sigma Delta Honorary Societies in 1934.

After returning to China he taught entomology and zoology first at the National Chekiang University, and later at the National Kwansi University. In 1936 he published the paper

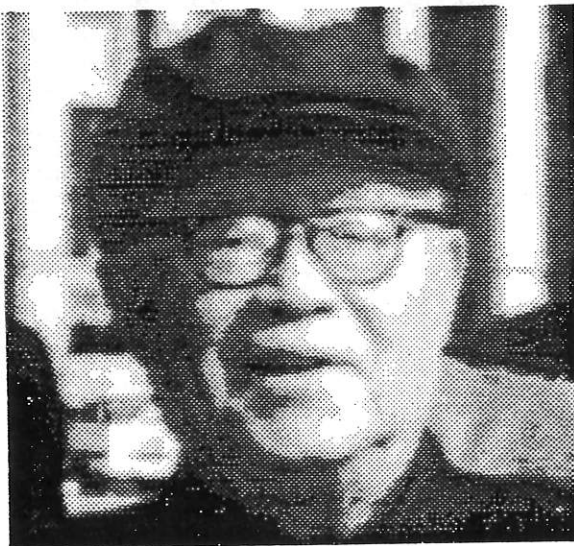
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"Checklist of Chinese Fleas" and in 1939 the treatise "The Fleas of China." The latter was the first monograph of the Siphonaptera of China and treated 74 species and subspecies. In the autumn of 1952 he joined the Institute of Microbiology and Epidemiology, Academy of Military Medical Sciences, Beijing, where he resumed his researches in medical entomology, particularly fleas.

Since 1973 he had edited the monograph "Order Siphonaptera, Fauna Sinica" under the auspices of the Editorial Committee, Academia Sinica. During this period, even at seven-

ty-plus years of age, he travelled the length and breadth of 14 provinces, visiting cities and autonomous regions, examining over ten thousand specimens, and consulting more than 500 references and



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books. Finally, the book was issued at the end of 1986 and dealt with 8 families, 72 genera and 452 species and subspecies, in addition to containing nearly 2000 figures.

Prof. Liu Chi-ying served on the editorial committees of Acta Entomologica Sinica and Acta Zootaxonomica Sinica and he was the vice director-general of the Entomological Society of China. He was an excellent teacher and tutor, modest and prudent, amiable and easily approachable, strict with himself and honest in performing his official duties. He gave no thought to personal gains or losses and was worthy of the name of teacher. He took an active interest in nurturing and educating young scientists and won praise from the masses.

Professor Liu Chi-ying contributed more than one hundred and fifty papers. Following is a bibliography of some of these studies dealing with flea taxonomy.

- 1935a On the unrecorded male of the bat-flea *Ischnopsyllus tateishii* Sugimoto. China J. Sci. Arts 23: 306-310, 2 figs.
- 1935b A new Chinese flea. Peking nat. Hist. Bull. 9: 273-275.
- 1936 Catalogue of Chinese Siphonaptera. Lingnan Sci. J. 15: 379-390, 583-594.
- 1939 The fleas of China. Order Siphonaptera. Philipp. J. Sci. 70: 1-122, 132 figs.
- 1956 Studies on the Chinese fleas. A description of the unrecorded female of *Nycteridopsylla galba* Dampf. Publ. Chin. PLA Inst. med. Sci. 2: 1-4.
- with Wu Hou-yong. 1960. Studies on the Chinese fleas, III. A new *Frontopsylla* from northwest China, a new *Ctenophthalmus* from southwest China, and a new *Stenoponia* from east China. Acta ent. sin. 10(2): 171-176, 2 pls.
- with Wu Hou-yong & Yu Jiu-fei. 1964. On a new species of the genus *Macrostylophora* (Siphonaptera: Ceratophyllidae). Acta zootax. sin. 1(1): 163-166.
- with Wu Hou-yong & Wang Dun-qing. 1965a. A new genus and species of flea, *Calceopsylla aduncata* gen. et sp. nov., from Tsinghai, China. Acta zootax. sin. 2(1): 57-62.
- with Wu Hou-yong & Wu Foo-lin. 1965b. A new species of *Amphipsylla* Wagner from west China (Siphonaptera: Leptopsyllidae). Acta zootax. sin. 2(4): 320-322.
- with Wu Hou-yong & Wu Foo-lin. 1965c. On two new species of *Vermipsylla* from west China and a revision of the characters of the genus (Siphonaptera: Vermipsyllidae). Acta zoo. sin. 17(4): 406-413.
- with Wu Hou-yong & Wu Foo-lin. 1966. Four new ceratophyllids from Tibet, China. Acta parasit. sin. 3(1): 64-71.
- with Tsai Li-yuen & Wu Wen-ching. 1974a. New genus and species of Siphonaptera from Chinghai province, west China. Acta ent. sin. 17(1): 102-111.
- with Wu Hou-yong & Chang Feng-po. 1974b. Notes on new Siphonaptera of Hopei province, north China. Acta ent. sin. 17(2): 219-223.
- with Wu Hou-yong & Wu Foo-lin. 1977. Discovery of *Sternopsylla* Jord. & Roths. in Tibet, China with descriptions of a new subgenus and three new species (Siphonaptera: Ischnopsyllidae). Acta ent. sin. 20(2): 229-235.
- with Wu Hou-yong. 1979a. A preliminary discussion on the faunal distribution and phylogeny of the fleas of China. Acta zootax. sin. 4(2): 99-112.
- with Zhung Zeng-hu & Wang Shi-xin. 1979b. Description of a new genus

- and species of Amphipsyllinae, Leptopsyllidae (Siphonaptera) from Kansu, China. Acta ent. sin. 22(2): 196-199.
- with Liu Yong-tai & Zhang-jun. 1980a. Description and discussion of a new genus and species of Ceratophyllidae from Shaanxi province, China. Entomotaxonomia 2(2): 135-139.
- with Cai Li-yun & Wu Wen-zhen. 1980b. Description and discussion on a new species of *Geusibia* from Qinghai province, China. Entomotaxonomia 2(3): 173-176.
- with Wu Hou-yong & Shao Guan-nan. 1980c. On *Nycteridopsylla* Oudemans, 1906 in China (Siphonaptera: Ischnopsyllidae). Entomotaxonomia 2(4): 291-296.
- with Liu Quan & Zhai Zhao-hua. 1981a. On a new species of *Macrostylophora* and notes on the genus in China. Entomotaxonomia 3(1): 23-27.
- with Hsing Chin-yun & Chen Jiu-min. 1981b. Description of a new species of *Ischnopsyllus* Westwood, 1833 (Siphonaptera: Ischnopsyllidae) from Shanxi, China. Acta ent. sin. 24(3): 317-320.
- with Chen Jia-xian & Liu Quan. 1981c. Description and discussion of a new species of *Geusibia* Jordan, 1932 from Sichuan (Siphonaptera: Leptopsyllidae). Entomotaxonomia 3(2): 95-98.
- with Zhang Di & Liu Quan. 1982a. Notes on *Megathoracipsylla pentagonia* Liu et al., 1980 and its unknown female (Siphonaptera: Ceratophyllidae). Entomotaxonomia 4(1-2): 37-38.
- with Liu Jiou & Liu Quan. 1982b. Description of a new species of *Chinghaipsylla* (Siphonaptera: Leptopsyllidae) with a discussion. Zool. Res. 3(Suppl.): 131-136.
- with Yu Xin & Ye Rui-yu. 1983. On the distribution of the subspecies *Ophthalmopsylla volgensis* and the description of a new subspecies in Xinjiang, China. Entomotaxonomia 5(2): 121-124.
- with Zhang Jin-tong & Wu Hou-yong. 1984a. Studies on Siphonaptera from Qinling Mountains, central China. I. Description of three new species and subspecies of Hystri-chopsyllidae. Acta zootax. sin. 9(3): 301-308.
- with Zhang Jin-tong. 1984b. Studies on Siphonaptera from Qinling Mountains, China. II. Description of new species and subspecies of *Geusibia* (Leptopsyllidae) and a preliminary approach to the systematic evolution of the genus. Acta zootax. sin. 9(4): 402-409.
- with Cai Li-yun & Wu Jia-song. 1984c. Description of a new species of *Callopsylla* Wagner, 1934 from Qinghai Province, China (Siphonaptera: Ceratophyllidae). Acta zootax. sin. 9(4): 410-412.
- with Zhang Jin-tong. 1985a. Studies on Siphonaptera from Qinling Mountains, China. III. Descriptions of two new species and one subspecies of Leptopsyllidae. Acta zootax. sin. 10(1): 60-66.
- with Cai Li-yun. 1985b. On a new species of *Frontopsylla* from Qinghai, China (Siphonaptera: Leptopsyllidae). Acta zootax. sin. 10(2): 184-186.
- with Liu Zheng-zhong & Zhang Jin-tong. 1986. A new subspecies of the genus *Paradoxopsyllus* Miyajima & Koidzumi, 1909 (Siphonaptera: Leptopsyllidae). Entomotaxonomia 8(1-2): 5-7.
- with Yu Xin & Ye Rui-yu. 1988. A new species of *Mesopsylla* (Siphonaptera: Leptopsyllidae). Entomotaxonomia 10(1-2): 1-4.



CONGRATULATIONS - DR. VLADMIR CERNY writes that he has retired as of January 1, 1989. Dr. Cerny was formerly a parasitologist associated with the Czechoslovakian Academy of Science and also actively involved with publication of the journal Folia Parasitologica. Readers will find his home address in the Mailing List section of this issue. The editors of Flea News wish him well in his retirement years.

TODD RAYMOND REICHARDT has completed work for the Master of Science degree at the University of Manitoba. His thesis is entitled "The seasonal dynamics and host-parasite relationship of *Opisocrostis bruneri* (Baker), a flea on Franklin's ground squirrel, *Spermophilus franklinii* (Sabine) near Birds Hill Park, Manitoba." Professor Terry D. Galloway supervised his studies.

THE SOCIETY FOR VECTOR ECOLOGY will hold its annual conference November 13, 14 and 15, 1989, at the University of Oklahoma, in Norman, Oklahoma. Further information may be obtained from B. Fred Beams, Chairman, Public Information Committee, Society for Vector Ecology, P.O. Box 87, Santa Ana, California, USA, 92702, telephone number (714) 971-2421, FAX number (714) 971-3940.

FLEA CELLS IN CULTURE. Persons knowing of researchers who have flea cells in culture are urged to contact Ms Beverly Dale of PARAVAX, 3181 Porter Drive, Palo Alto, California 94022, (415) 493-1693.

WARNING!

In the United States, one of the most popular floor coverings is DuPont's Stainmaster® carpet. A note in the November issue of Pest Control magazine indicates that the application of a pesticide to this type of carpet will void the warranty. Although not publicized, DuPont takes the position that pesticides interfere with the chemical used to treat the carpet yarn for stain resistance. This would

include any chemical which might be used to control fleas in the home. Whether methoprene and other insect growth regulators (IGR) are also included is not indicated, but they probably are.

PLAGUE UPDATE. In early January, the Brazilian Ministry of Health notified WHO of 8 non-fatal cases of plague from central Bahia State east of the San Francisco River. The report covered the period 19 January through 11 July 1988.

According to unofficial reports, plague is reemerging in Yunnan, Gansu and Heilongjiang Provinces, and Xinjiang Uygur and Inner Mongolia Autonomous Regions of the People's Republic of China. Delayed reports to WHO list 7 cases, 2 of which were fatal, during 1987. No provinces were specified but Xinjiang and Qinghai Provinces are known to contain endemic foci.

Recent serosurveys conducted in Rafah and El Arish, Egypt, revealed that 23% of the human sera were positive for antibodies of *Rickettsia typhi*, the causative agent of murine (fleaborne) typhus. This disease may be a significant cause of febrile illness in the northeastern Sinai Peninsula.

Closer to home, the first case of plague in the United States for 1989 was reported to the Communicable Disease Center, Fort Collins, Colorado on 12 June, involving a 47 year old hispanic male from Rio Arriba county, New Mexico. Since then an additional case has been confirmed in a 14 year old male from southern Colorado.



Dr. R.L.C. PILGRIM has contributed the following account of his experiences while attending the Bratislava Symposium in the summer of 1988. A list of the papers can be found in the PAPERS PRESENTED section of this issue.

The Symposium entitled "The Results and Perspectives of Further Research of Siphonaptera in Palearct from the Aspect of their Significance for Practice" was held in Bratislava, Czechoslovakia, from 6 - 11 June, 1988. It was sponsored by the Slovak Entomological Society of the Slovak Academy of Sciences and the Department of Zoology, Comenius University, under the immediate chairmanship of Prof. Dr. M. Kiefer. Meetings took place in the University's Congress Hall.

About 40 participants were present, the majority coming from the Soviet Union, and from the host country, with small numbers from Poland, Bulgaria, Romania, and East Germany. I was not only the sole representative from New Zealand, but the only member from "the west!" As such, I was fêted and given V.I.P. treatment, including being allocated two charming young Slovak students to act as my interpreters for the week. In fact, these young women, though gracious and very helpful, were less essential than I expected, as most of the participants spoke English -- some of them very well indeed -- and all appeared to be able to understand it readily. Many of the conference contributions were given in Russian, and my interpreters were very valuable then, in giving me a running commentary on the talks. My own contributions were, however, made in English without simultaneous translation and most of the audience seemed to understand well.

Most of the contributions were devoted to aspects of adult flea research: they

ranged from the origin of fleas as deduced from the fossil record, the origin and composition of Palaearctic flea fauna, to stochastic modelling of flea-host populations, and some purely taxonomic studies.

A large proportion of the papers presented related to the flea fauna of the Asian far east, and especially to Mongolia; the Slovak Academy of Sciences has co-operated for a number of years with the Soviet Academy of Sciences in expeditions to that part of Asia, and the two institutions have made a number of joint publications as a result. The collaboration is on-going and many interesting results were forthcoming at the conference.

For me, the highlight of the conference was an afternoon session, followed by much of the next morning, devoted to larval studies. Papers were presented on keys to identify larvae of several families (Pulicidae, Ctenophthalmidae, Hystri-chopsyllidae, Leptopsyllidae, Ceratophyllidae) of the Palaearctic region. My paper was a more general one, devoted to external morphology of flea larvae and the use of scanning electron microscopy in re-evaluating interpretations made with the optical microscope. The ensuing discussions at this session were the most protracted and stimulating of the conference and provided a very useful meeting ground in this discipline of workers from centres remote from each other. Good contacts were made, and some very rewarding friendships established. The overall impressions are of a most successful and entertaining international gathering.

Proceedings of this Symposium are expected to be published in the near future, under the general editorship of Dr. Matej Kiefer.

- RLCP



PAPERS PRESENTED

The following presentations dealing with fleas were delivered at the annual conference of the ENTOMOLOGICAL SOCIETY OF AMERICA held in Louisville, Kentucky, December 4 - 8, 1988.

Kim, K. C., K. C. Emerson & R. Traub. Diversity of parasitic insects -- Anoplura, Mallophaga and Siphonaptera.

Hinkle, N. C., R. W. Wadleigh, P. G. Koehler & R. S. Patterson. Comparison of acetylcholinesterase activity and inhibition in resistant and susceptible cat fleas (*Ctenocephalides felis*).

Silapanuntakul, S. & F. W. Knapp. *In vitro* feeding *Ctenocephalides felis* through a reinforced silicone membrane.

Palma, K. & R. Meola. Residual activity of Sumitomo S-31183 against cat flea larvae in Texas yards.

Silapanuntakul, S. & F. W. Knapp. An artificial feeding *Ctenocephalides felis* through silicone membrane.

The European Region of the Society for Vector Ecology held its annual conference August 30 to September 2, 1988, at Pembroke College, University of Cambridge, Cambridge, England. One paper dealing with fleas was presented:

"Use of Indicators of Sexual Maturation and Regression in Fleas in Host-Specificity Studies" by B. Williams, University of Cambridge, Dept. of Applied Biology [now with Dept. of Zoology, see Mailing List section].

The Department of Zoology, Faculty of Natural Sciences, Comenius University, Bratislava, Czechoslovakia, sponsored a symposium entitled "The results and perspectives of further research of Siphonaptera in [the] Palaearctic from the aspect

of their significance for practice," June 6-11, 1988. A volume of English abstracts has since been prepared and following is a list of the presentations.

Ponomarenko, A. G. The Origin of fleas: the evidence from fossil record.

Zhovtyi, I. F. Review of the perspectives in studies of Siberia and the Far East.

Krylova, T. V., G. A. Askerov & N. S. Agaeva. Spatial distribution of fleas/Siphonaptera in local populations of Meriones Libycus Licht.

Klimaszewski, S. M. & T. V. Krylova. Consortial and correlative relations within flea/Siphonaptera/communities on small mammals.

Jedlicka, L. & M. Kiefer. Probability distribution models for some flea species of Siberian Jerboa, *Allactaga sibirica* Forster.

Ryba, J., D. Cyprich & J. Minar. Fleas on domestic and straying dogs.

Goncharov, A. I., M. Kiefer & V. S. Lobachev. The formation of the fleas fauna of the Palearctic.

Serzhanov, O. S., M. Kiefer & V. S. Lobachev. Faunistic relations with Palearctic and Nearctic on mammalian fleas as an example/except Chiroptera/.

Poslavskii, A. N. On the history of fauna formation in the northern deserts of the USSR.

Romasheva, T. P. Contemporary studies of flea fauna and geography in the Far East.

Nazarova, I. Formation of Aphaniptera of eastern Europe.

Dudich, A., L. Cristov & P. Slavova. Sup-

plements to the fauna of Bulgaria and preliminary check-list.

Buschueva, L. & O. Gorbunov. New data on fleas/Siphonaptera/of central Changaj in Mongolia.

Medvedev, S. G. Revision of the family Ischnopsyllidae/Siphonaptera/.

Goncharov, A. I., M. Kiefer & V. S. Lobachev. On fleas of Genus Chaetopsylla.

Vashchonok, V. S. Physiological and morphological peculiarities of fleas/Siphonaptera/as vectors of disease agents.

Teplykh, V. S., I. A. Yurgenson & G. O. Lilp. Preimaginal diapause in fleas/Siphonaptera/.

Rödl, P. The effect of host hibernation on the behavior of fleas *Ceratophyllus gallinae*.

Dudich, A. & J. Kleinert. Pitfall traps as a method of collecting of small mammal fleas: its advantages and limits.

Goncharov, A. I., M. Kiefer & V. S. Lobachev. Recommendation on identification of some subspecies of fleas in Mongolia.

Dmitriev, P. P., M. Kiefer & A. A. Tarakanovski. The set of flea species as indicator of different populations of the Brandt's vole in Mongolia.

Suntzova, N. & V. Suntzov. Distribution of fleas/Siphonaptera/in populations of the longtailed suslik.

Tshikhistavi, Sh. G., A. I. Goncharov, L. I. Belyavtseva, A. A. Goncharov, K. I. Arakoshvili, R. A. Ghasanov & J. A. Safarova. On ectoparasites of *Chionomys* and *Microtus* from the Caucasus.

Suntzova, N. & V. Suntzov. Distribution

of fleas/Siphonaptera/in the Microhabitat of *Ochotona Pricei* Thom.

Cerny, V. Fleas and phoretic hypopi of mites.

Pilgrim, Robert L. C. Flea larvae-their morphology in relation to taxonomy, biology and phylogeny.

Goncharov, A. I., M. Kiefer & P. C. Karandina. Table for identification of larvae of investigated genera of Ctenophthalmidae, Hystrichopsyllidae and Leptopsyllidae.

Máca, J. Notes on the determination of the flea larvae.

Goncharov, A. I. Table for identification of larvae of investigated genera of Pulicidae.

Goncharov, A. I., M. Kiefer & P. C. Karandina. Table for identification of the second and third age of some genera of Ceratophyllidae according to their larvae.

Suciu, M. & F. Aioanei. Some taxonomical and zoogeographical aspects of the *Rhadinopsylla* Jordan and Rothschild/Insecta, Siphonaptera/occurring in Romania in the general background of the European Palearctic.



RECENTLY RECEIVED

The jerboa fauna of the SSSR edited by A. N. Severtsova and published by the All-Union Thierological Society of the SSSR, Institute of Epidemiology & Microbiology, N. F. Gamalei Academy of Medical Sciences of the SSSR. 246 pp. (1985), Moskva. The work deals with the phylogeny, systematics, faunistics, morphology,

ecology and behavior of jerboas, and in a section entitled "Biological Communications," the following papers dealing with fleas appear.

Z. P. Vakhrusheva. Concerning contact with parasites in a natural plague focus in the Zabaikal region. p. 187-189.

A. I. Goncharov. About fleas parasitic on jerboas. p. 189-194.

I. Zh. Zhubanazarov, L. A. Burdelov, T. A. Tyurina, L. A. Dolotova, L. I. Kockkina, H. Yu. Rudenchik, V. A. Burdelov, N. V. Burdelova & P. G. Skvortsova. Jerboa fleas in the Priaral region. p. 194-197.

V. C. Lobachev, M. Kiefer & M. Krumpal. Jerboa fleas from Mongolia. p. 206-207.

T. V. Krylova & M. Kiefer. The fleas of *Allactaga sibirica* in the Yuzhnykh region of Tuva. p. 208-209.

O. V. Mitropol'skiy, A. V. Polyanskiy & T. A. Nikitiana. The fleas of *Jaculus turcomenicus* and *Eremodippus* [= *Jaculus*] *lichtensteini* in the Kyzil-Kum region. p. 209-212.

N. V. Ol'kova. The epizootological role of *Allataga sibirica* in the Siberian plague focus. p. 212-214.

V. N. Prokop'ev, A. G. Stupina, R. T. Maturova & T. P. Nikhileeva. Ectoparasites of *Allactaga sibirica* in the south-western Zabaikal region. p. 214-215.

E. G. Samarin, P. N. Kiselev & F. A. Saraev. Some notes on jerboa ecology in the north-eastern Pricaspian region, and their fate in the epizootology of plague. p. 215-219.

O. S. Serzhanov. Zoogeography, genetic analysis and distribution of Palaearctic jerboa fleas. p. 220-226.

O. S. Serzhanov, N. T. Kunitskaya & Z. P. Maslennikova. Jerboa fleas in Central Asia and Kazakhstan. p. 227-228.

E. G. Samarin, L. M. Samarina, V. I. Il'yushkina & F. A. Saraev. Jerboa fleas in the northeastern Pricaspian region. p. 229-233.

A. E. Chegodaev. On the role of jerboas in the eastern Zavcaucasus region in the epizootic of plague. p. 233-234.

V. L. Shevchenko. On the relationship of the jerboas of the stony-deserts of the northern Pricaspian region and the natural focus of plague. p. 235-237.

V. I. Yurkina. Some ecological peculiarities of jerboa fleas in the Ukraine. p. 240-242.

REPRINTS

Since Flea News 37, we have received reprints and photopies from:

Benton, A. Cerny, V., Cyprich, D., Dryden, M., Dudich, A., Egoscue, H., Georgi, J., Gomez, M., Greeve, J., Hesse, G., Kiefer, M., Lane, R., Larson, O., Li K. -c., Medvedev, S., Prasad, R., Ribeiro, H., Sapegina, V., Saxena, V., Skuratowicz, W., Smith, S., Stanko, M., Xie, B. -q.

We are grateful for these additions to the Flea News library.



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DELETIONS

Dr. J. Abul-hab, Prof. Dr. S. S. Ahmed,
Dr. A. N. Alekseyev, Eric Anderson, Da-
vid L. Bishop, Mrs. G. Brinck-Lindroth,
David Broadbent, Ms Page Burt, Dr. L.
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Enrique R. Justo, Dr. J. M. Klein, N. L.
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Lloyd, Henry L. Martin, C. Maser, Dr.
John Matteson, Brendan C. McKiernan,
Dr. Bruce S. Ott, Peter-Paul, Dr. Rita H.

Rijntjes, R. E. Ryckman, Dr. Y. Schlein,
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Singh, Dr. E. F. Sosnina, Ruth Sponsler,
Dr. JoAnn Tenorio, Mr. Robert F. Trigg,
Michelle Turnbull, Juha Vuorimies, Jack
Waldheim



ADDENDA

Your attention is called to the descriptions of two additional taxa previously omitted from our lists. To the list for 1982 (Flea News 29, p. 144) please add:

98230 sichuanxizangensis Wu & Chen
Neopsylla specialis

The citation for this paper is:

Wu H. -y. & Chen N. -y. 1982. Description of a new subspecies of Neopsylla Jordan (Siphonaptera: Hystrichopsyllidae). Sichuan J. Zool. 2: 27-29.

To the species list for 1986 (Flea News 36, p. 279) please add:

98632 hongyangensis Li, Bai & Chen
Neopsylla

The citation is:

Li K. -c., Bai X. -l. & Chen B. -f. 1986. A new species of the genus Neopsylla from Ningxia. Sichuan J. Zool. 5: 1-3.

A PREFATORY NOTE TO THE SPECIES LIST -- In keeping with Mr. Smit's practice of numbering the taxa chronologically, the following list includes numbers for those taxa described in 1987. Readers are urged to call any omissions in these lists to our attention.

98701 janineae Beaucournu & Gouat
Caenopsylla

98702 ibera Beaucournu & Marquez
Caenopsylla laptevi

98703 gongheensis Zhang & Ma
Amphipsylla longispina

98704 xiei Li Stenischia

98705 viracocha Schramm & Lewis
Plocopsylla

98706 nungui Schramm & Lewis
Plocopsylla

98707 athena Schramm & Lewis
Plocopsylla

98708 kilya Schramm & Lewis
Plocopsylla

98709 intermedia Cai, Wu & Zhang
Frontopsylla (Frontopsylla)
elatoides

98710 falsotorosa Wu, Ni & Li Geusibia

** 98711 hamutus Cai & Wu
Certophyllus (Monopsyllus)

98712 forficus Cai & Wu Ceratophyllus
(Monopsyllus)

98713 nandanensis Li, Zeng & Zeng
Macrostylophora

98714 tianshanensis Yu, Ye & Xie
Amphalius clarus

98715 nomisis Smit Ectinorus
(Ectinorus)

98716 deplexus Smit Ectinorus
(Ichyonus) onychius

98717 ambrosius Smit Parapsyllus

98718 deflatus Smit Tiamastus

98719 rangeli Smit Polygenis
(Gephyropsylla) klagesi

- 98720 *mendezi* Smit Polygenis
(Polygenis) *trapidoi*
- 98721 *rhombomysus* Li, Huang & Sun
Paradoxopsyllus
- 98722 *inferioprocerus* Yu, Cao & Huang
Paradoxopsyllus
- 98723 *liui* Yu *Wagnerina*

** The specific name hamutus was misspelled previously in *Flea News* as hamatus: FN 36, p. 282, last line from the bottom; FN 37, p. 297, line 12. *Mea culpa*.
- J.H.L.

PROVISIONAL LIST OF TAXA DESCRIBED IN 1988

- Amphipsylla apiciflata* Liu, Xu & Li
- Paradoxopsyllus aculeolatus* Ge & Ma
- Amphipsylla yadongensis* Wang & Wang
- Plocopsylla lewisi* Beaucournu & Gallardo
- Ctenoparia propinqua* Beaucournu & Gallardo
- Nycteridopsylla liui* Wu, Chen & Liu
- Callopsylla arcuata* Ge, Wang & Ma
- Metastivalius novaehiberniae* Beaucournu & Mahnert
- Mesopsylla sagitta* Yu, Ye & Liu
- Ischnopsyllus* (*Hexactenopsylla*) *infratentus* Wu, Wang & Liu
- Pulicella aenigma* Lewis & Cheetham
- Plocopsylla kasogonaga* Schramm & Lewis

Macrostylophora angustihamula Li, Zhang & Zeng

Ctenophthalmus longiprojiciens Chen, Li & Wei

PROVISIONAL LIST OF TAXA DESCRIBED IN 1989

- Palaeopsylla laxidigita* Xie & Gong
- Palaeopsylla polyspina* Xie & Gong
- Palaeopsylla medimina* Xie & Gong
- Ctenophthalmus* (*Palaeoctenophthalmus*) *harputus* Aktas



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1985 (List 8)

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